Intro to Software Engineering

Final Project Report

NUDTEE

A real-time web application for scheduling group activities and tracking headcount

Team Dek70

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Repository: https://github.com/6931503074-sudo/NUDTEE1

Prototype / demo: https://nudtee-e329d.web.app/

1. Problem and Target Users

NUDTEE is a web application that streamlines general activity scheduling and headcount tracking in real time. Organising activities through LINE or Facebook group chats creates seven documented problems:

1. Activities are announced informally in chat threads, with no structured place for exact details, schedules and locations.
2. Organisers have no reliable running headcount and must ask around and count RSVPs by hand.
3. Headcounts discussed in chat go stale as people change their minds.
4. Unclear or vague locations cause participants to get lost or arrive late.
5. Logistics are discussed across several general chat groups and get lost or mixed up.
6. Last-minute joins after preparations are finalised cause headcount mismatches.
7. Participants forget the date and time of an activity when the schedule exists only in chat threads.

The target users are participants who look for activity sessions with clear dates, times and locations, and session organisers who need to know how many people have joined without recounting RSVPs in chat.

Success measure and project scope (M1):

ID	Metric	Target
SM-1	Time for an organizer to create a session	60 seconds or less
SM-2	Taps for a player to join an open session from the list	2 taps or fewer
SM-3	Time for another device to show a join/leave change	2 seconds or less
SM-4	Taps to add a joined session to Google Calendar (excluding Google sign-in)	2 taps or fewer
SM-5	Rejection rate for join attempts (through the application) after registration has closed	100%
SM-6	Delivery time for in-room chat messages across devices	2 seconds or less

2. Requirements Summary

Functional requirements and priorities

ID	Requirement	Priority
FR-1	Create a session (title, location or online meeting link, date/time, minimum/maximum players, registration deadline, category).	Must
FR-2	Join or leave a session, with a live remaining-spot count.	Must
FR-3	Display real-time session status.	Must
FR-4	Display a session’s location on an interactive map.	Must
FR-5	In-room chat for session participants.	Must
FR-6	Auto-close a session when registration closes.	Must
FR-7	Add a joined session to Google Calendar.	Must

Non-functional requirements and acceptance criteria:

ID	Kind	Statement (testable)	How we will check
NFR-1	performance	The session list shall reflect a join/leave change, and in-room chat messages shall be delivered, to all connected clients within 2 seconds.	Two devices open the same session/chat; one joins/sends a message, time until the other updates/receives it.
NFR-2	usability	The app shall be usable on a mobile phone screen without horizontal scrolling on any core screen. Adding a session to Google Calendar, and joining an open session from the session list, shall each take 2 taps or fewer (excluding Google sign-in).	Manual check on a real phone for each of the core screens; manually count the taps for the calendar export and for joining from the list.
NFR-3	capacity	The system shall support at least 20 concurrently connected users viewing the real-time session list and sending chat messages without errors.	Load test with 20 simulated concurrent connections interacting with real-time listeners.
NFR-4	reliability	A session's remaining-spot count shall never go negative or exceed the stated maximum. 100% of join attempts submitted through the application after registration has closed shall be rejected.	Two clients tap Join simultaneously on a session with one spot left; verify only one succeeds. Attempt to join a closed session and verify immediate rejection.
NFR-5	security / privacy (required)	A user shall be able to edit or close only sessions they created, and cannot read the in-room chat unless	Attempt to modify another account's session directly through

ID	Kind	Statement (testable)	How we will check
		they are a participant. These rules shall be enforced by the database security rules, not only by the user interface.	the database API and confirm it is rejected. Attempt to fetch chat messages of an unjoined session and confirm denial.
NFR-6	usability	An organizer shall be able to create a session with all required fields in 60 seconds or less.	Time 3 test users creating a session from the home screen with a stopwatch.

3. System Design and Diagrams

3.1 Architecture

Three-tier architecture

Tier	Components	Requirement
Presentation	Session list screen (FR-1, FR-3, FR-6): open sessions with category, location, time, status (Open, Almost Full, Full, Joined, Closed) and a live remaining-spot count; category filter. Create session screen (FR-1): form with place search, current-location picker, online option, player limits and registration deadline. Session details screen (FR-2, FR-4, FR-7): details, map preview and View Location, Join / Leave, Add to Calendar. Room chat screen (FR-5): messages for joined participants.	FR-1 to FR-7

Tier	Components	Requirement
Application logic	Create session (FR-1): validates fields, limits, dates and time overlap; stores the session and adds the organiser as first participant. Join session (FR-2, FR-6): checks registration deadline and time overlap, then runs a Firestore transaction that re-reads the participant list and rejects the join when full. Leave session (FR-2): removes the participant and recalculates remaining spots. Evaluate session status (FR-3, FR-6): computes Open, Almost Full, Full, Joined or Closed when the list or details are displayed. Real-time synchronisation (FR-3, FR-5): Cloud Firestore listeners on session and chat data; they resume after a connection loss. Chat access check (FR-5): only participants may open or send messages. Build calendar link (FR-7): pre-filled Google Calendar event-template link (Asia/Bangkok).	FR-1, FR-2, FR-3, FR-5, FR-6, FR-7
Data	Cloud Firestore (Spark plan) and Firebase Authentication (Google sign-in): User, Session, Participation and Message data (Section 3.2). Database security rules enforce who may close a session or read a chat.	FR-1 to FR-7; NFR-5

3.2 Data Model

Entities, fields and relationships

Entity	Fields	Relationships	Used by
User	Google account ID, display name; optional username, short bio and profile picture (the full profile is kept in the browser's local storage).	Organises many Sessions; has many Participations; sends many Messages	FR-1, FR-2, FR-5
Session	id, organiser (User), title, short and room descriptions, category, online/offline type, location name, address and coordinates (or meeting link), date, start/end time, minimum and maximum players, registration deadline, cover image	Belongs to one organising User; has many Participations and Messages	FR-1, FR-3, FR-4, FR-6, FR-7
Participation	session (Session), user (User), display name and small picture thumbnail copied from the user's profile (stored with each session)	Belongs to one Session and one User	FR-2, FR-3, FR-5
Message	session (Session), sender ID, sender name, text, timestamp	Belongs to one Session and one User (sender)	FR-5

3.3 Diagrams

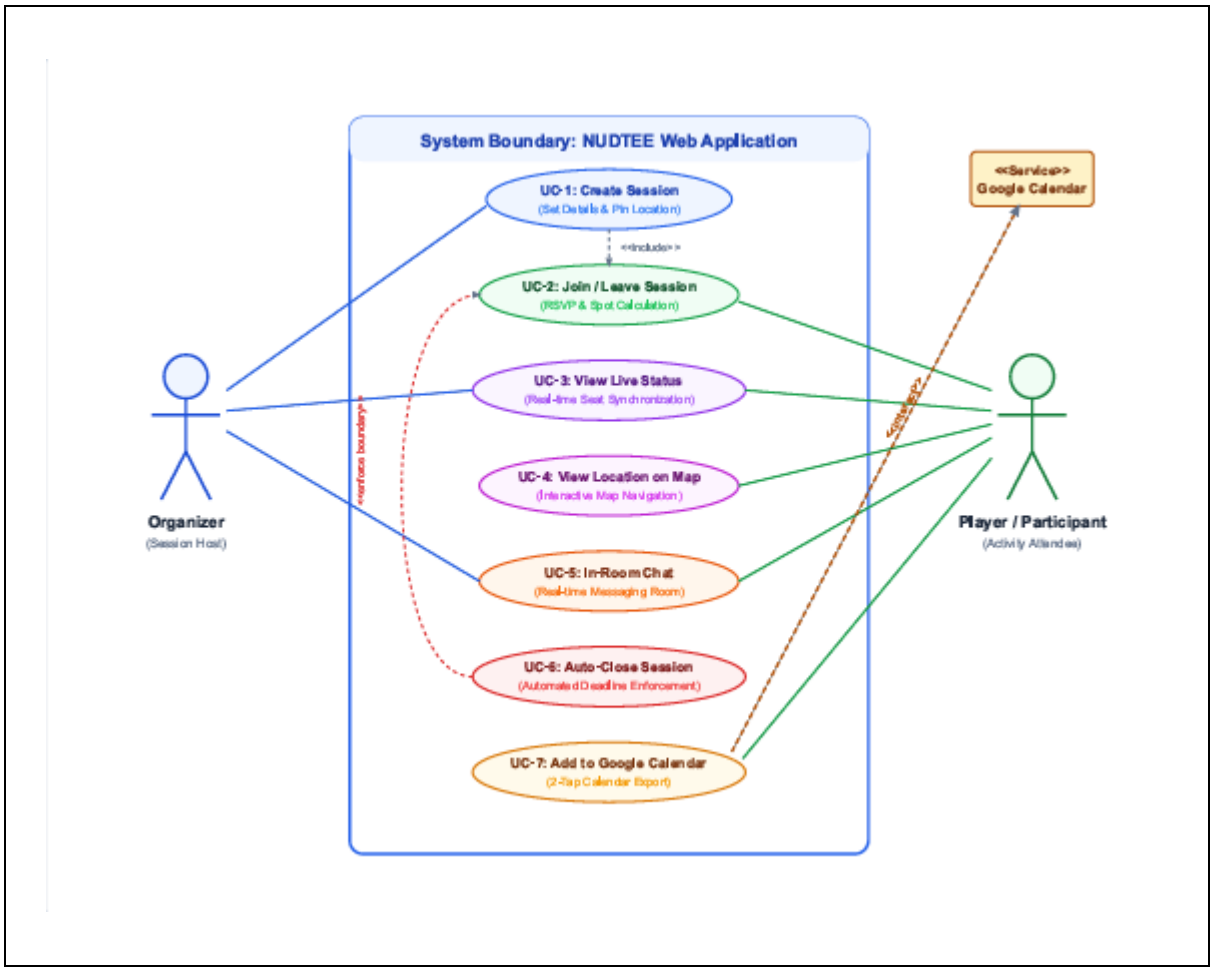


Figure 1. Use Case diagram of NUDTEE

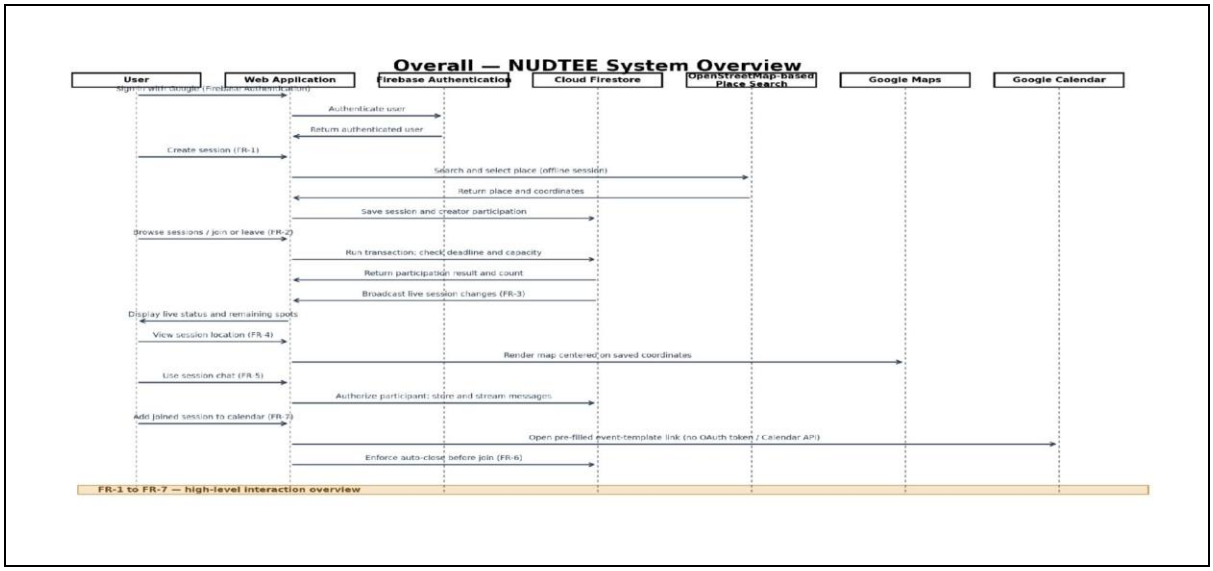
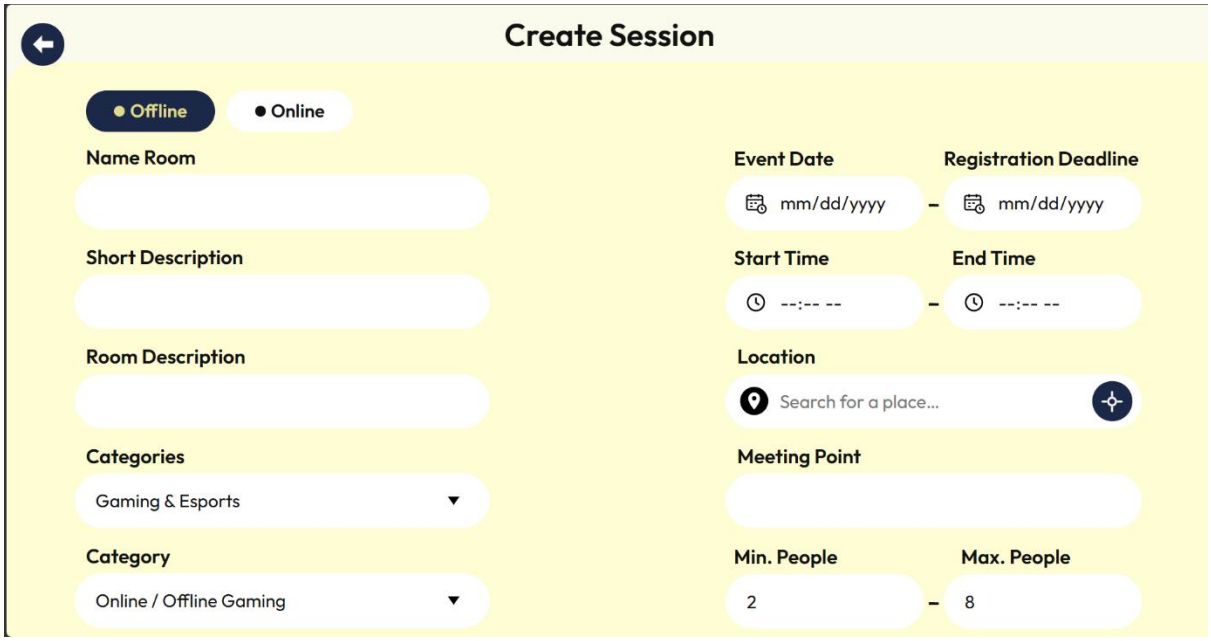
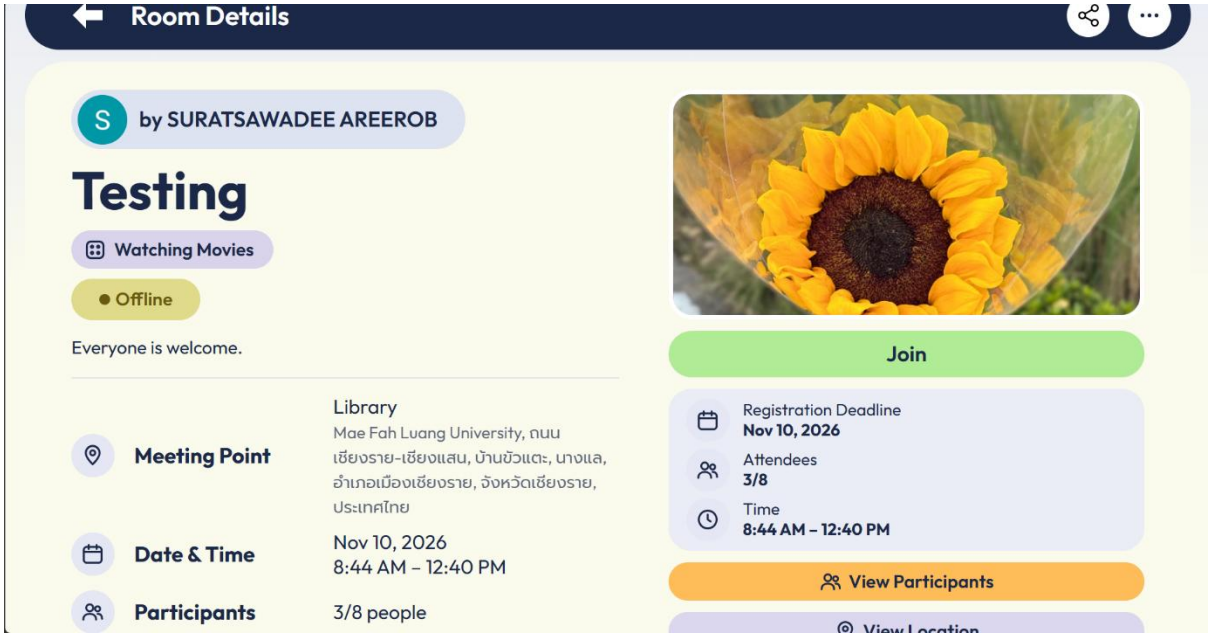


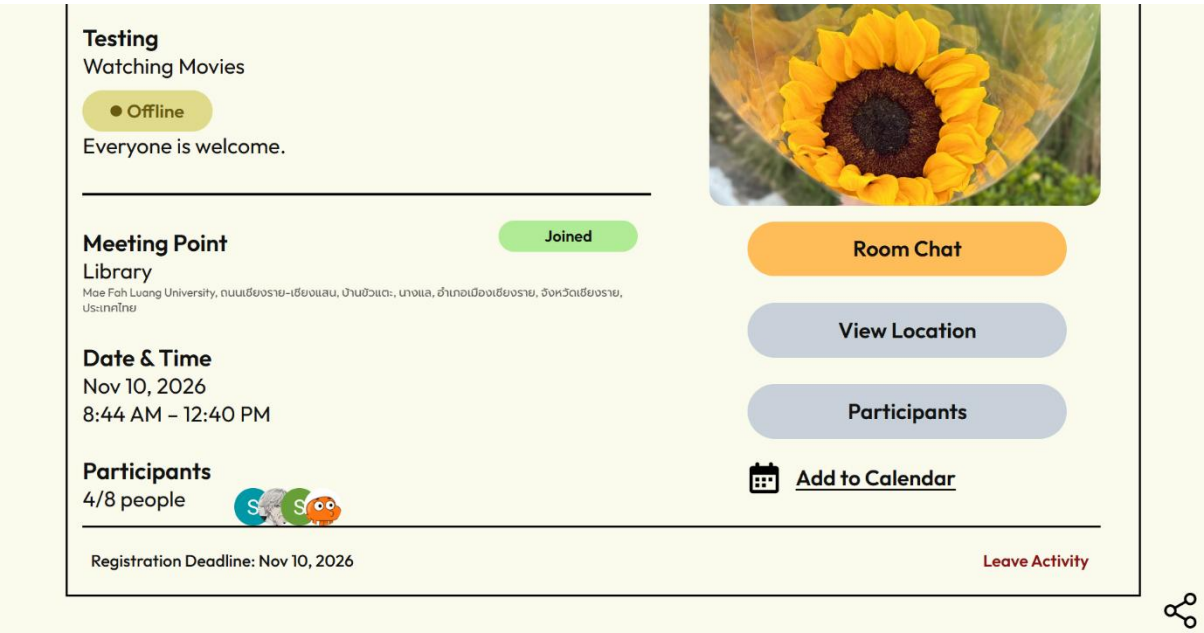
Figure 2. Sequence diagram of NUDTEE

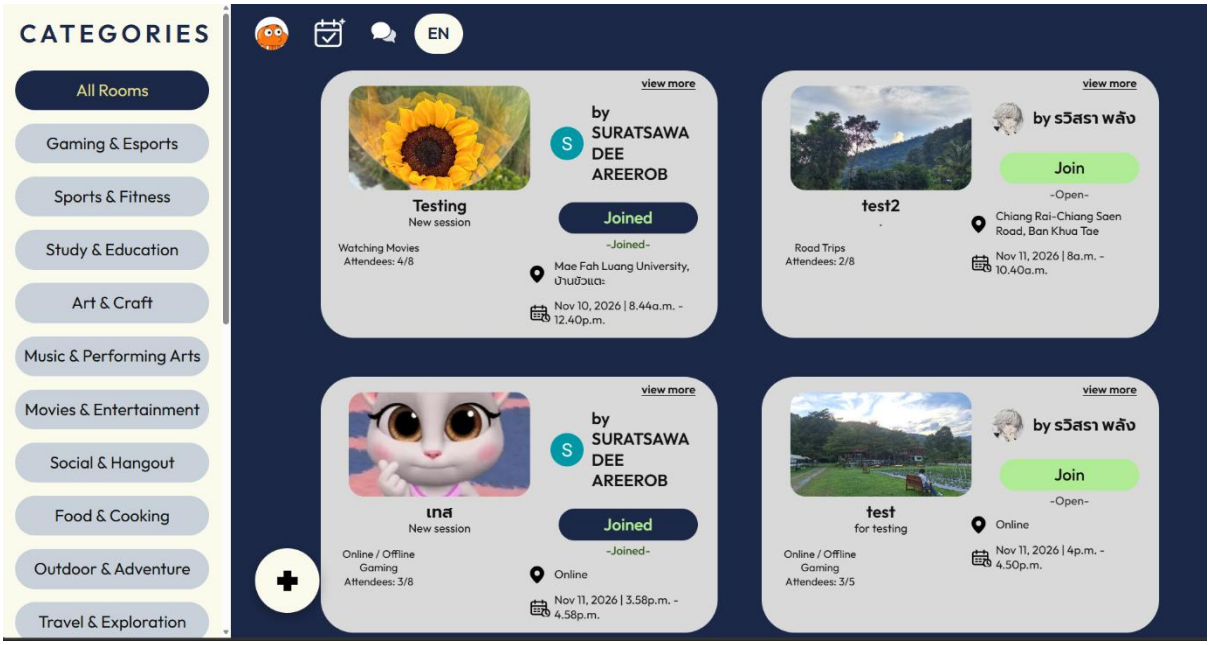
4. Prototype and Implementation

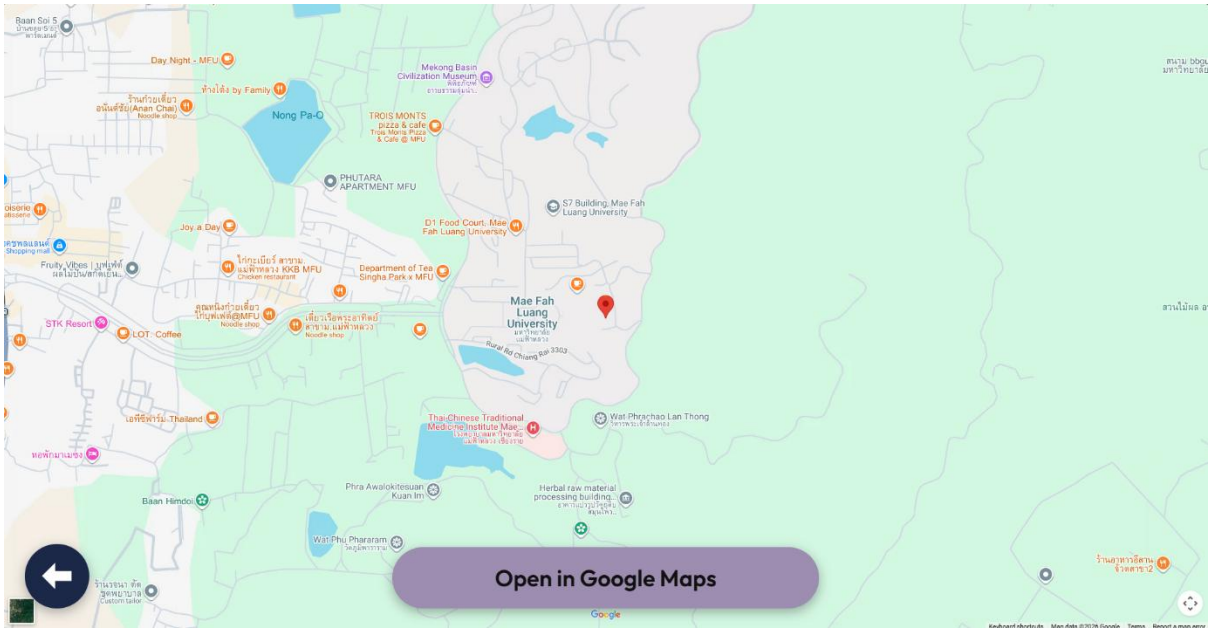
Prototype:

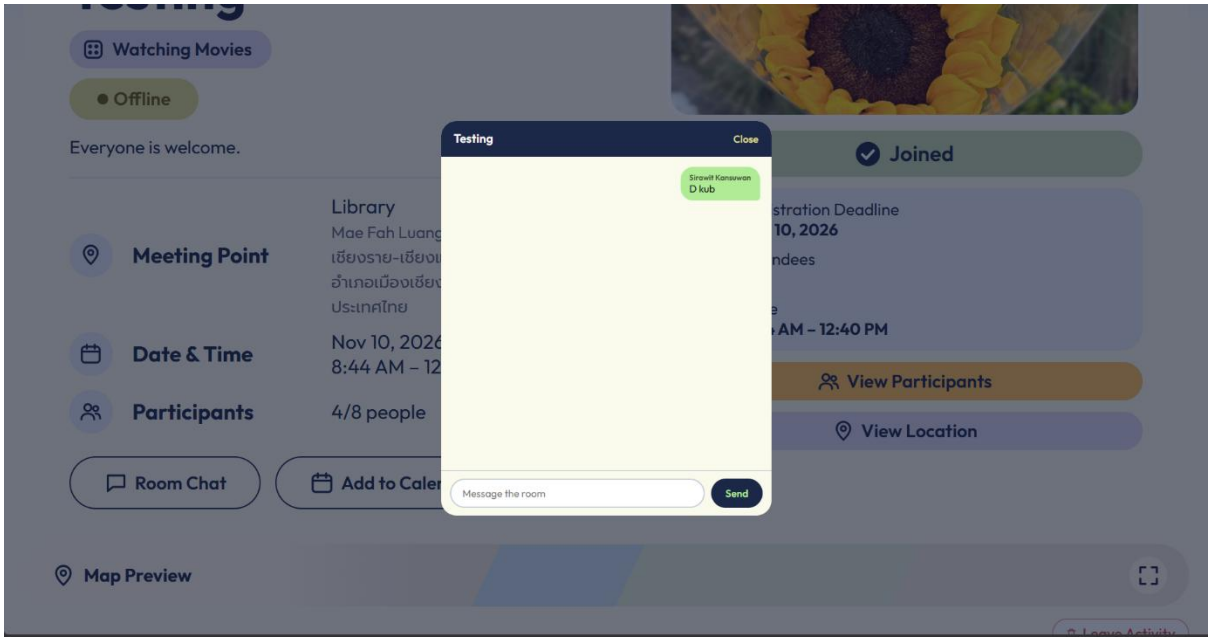


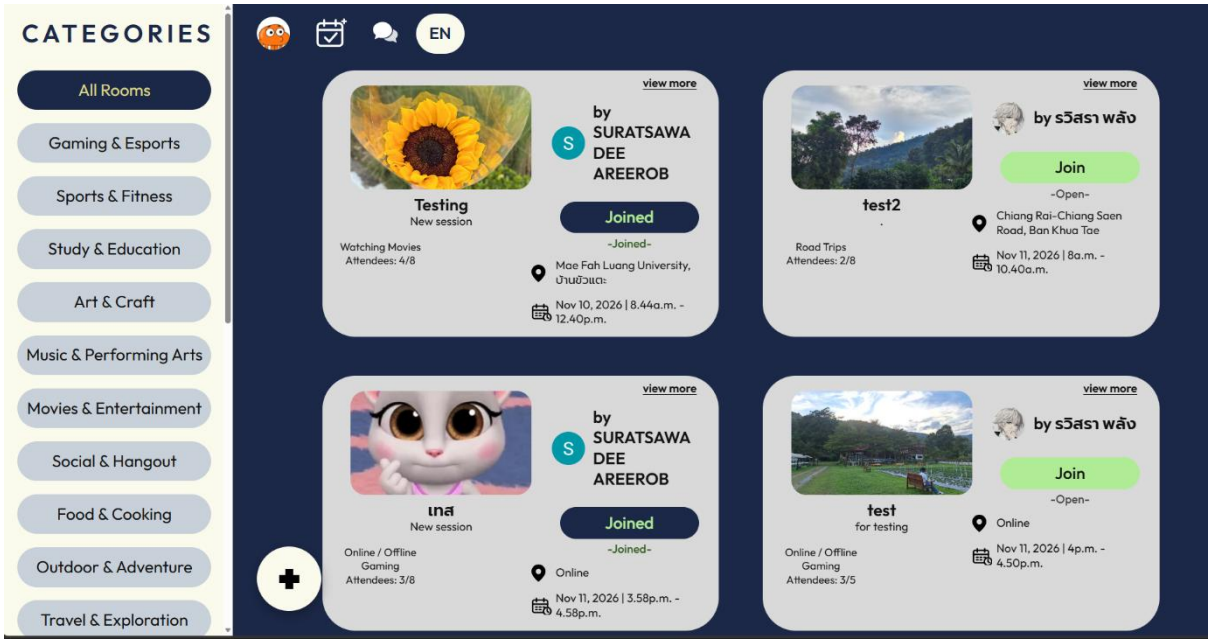



Testing
Watching Movies
Offline
Everyone is welcome.
Meeting Point
Joined
Library
Mae Fah Luang University, ถนนเชียงราย-เชียงแสน, บ้านขัวแตะ, นางแล, อำเภอเมืองเชียงราย, จังหวัดเชียงราย, ประเทศไทย
Date & Time
Nov 10, 2026
8:44 AM – 12:40 PM
Participants
4/8 people
Registration Deadline: Nov 10, 2026
Room Chat
View Location
Participants
Add to Calendar
Leave Activity


CATEGORIES
All Rooms
Gaming & Esports
Sports & Fitness
Study & Education
Art & Craft
Music & Performing Arts
Movies & Entertainment
Social & Hangout
Food & Cooking
Outdoor & Adventure
Travel & Exploration
EN
view more
Testing
New session
Watching Movies
Attendees: 4/8
by SURATSAWA DEE AREEROB
Joined
-Joined-
Mae Fah Luang University, บ้านขัวแตะ
Nov 10, 2026 | 8.44a.m. - 12.40p.m.
test2
Road Trips
Attendees: 2/8
by รวิสรา พลัง
Join
-Open-
Chiang Rai-Chiang Saen Road, Ban Khua Tae
Nov 11, 2026 | 8a.m. - 10.40a.m.
เทส
New session
Online / Offline Gaming
Attendees: 3/8
Online
Nov 11, 2026 | 3.58p.m. - 4.58p.m.
test
for testing
Online / Offline Gaming
Attendees: 3/5
Online
Nov 11, 2026 | 4p.m. - 4.50p.m.


Nong Pa-O
PHUTARA APARTMENT MFU
Mae Fah Luang University
Open in Google Maps


Watching Movies
Offline
Everyone is welcome.
Meeting Point
Library
Date & Time
Participants
4/8 people
Room Chat
Map Preview
Testing
Close
Sirawit Kansuwon
D kub
Message the room
Send
Joined
View Participants
View Location



Status of each requirement

ID	Planned (SRS)	Designed	Prototype	Implemented	Tested
FR-1	Yes	Yes	Yes	Yes	Yes
FR-2	Yes	Yes	Yes	Yes	Yes
FR-3	Yes	Yes	Yes	Yes	Yes
FR-4	Yes	Yes	Yes	Yes	Yes
FR-5	Yes	Yes	Yes	Yes	Yes
FR-6	Yes	Yes	Yes	Yes	Yes
FR-7	Yes	Yes	Yes	Yes	Yes

5. Testing and Results

Planned test scenarios and results

Test	FR	Scenario (planned)	Expected result (from design)	Actual result
TC-1	FR-1	An organizer submits a valid new offline session (location chosen from the place-search results).	The session is saved, appears in the session list under its category, its details page shows the details, and the organizer is added as the first participant.	Pass
TC-2	FR-1	An organizer saves with a required field empty, or with a typed location not chosen from the search results.	The save is rejected and an error message appears (“Pick a real place from the search results.” for the location).	Pass
TC-3	FR-1	An organizer saves with min players below 1, max below min, a deadline after the event date, or an end time not after the start time.	The save is rejected with an error message.	Pass
TC-4	FR-2	A player taps Join on an open session with remaining spots.	The player is added to the participant list and remaining spots decrease by 1.	Pass

Test	FR	Scenario (planned)	Expected result (from design)	Actual result
TC-5	FR-2	A player tries to join a session that is full.	The join is rejected and “This room is full.” is displayed.	Pass
TC-6	FR-2	Two players tap Join at the same time with only 1 spot left.	Exactly one succeeds; the other receives “This room is full.”	Pass
TC-7	FR-2	A participant who is not the organizer taps Leave.	The participant is removed and remaining spots increase by 1.	Pass
TC-8	FR-3	One device joins or leaves a session while another device views the session list.	The other device shows the updated count within 2 seconds without refreshing.	Pass
TC-9	FR-3	A client loses its connection and then reconnects.	The real-time listeners resume automatically and the displayed data resynchronises.	Pass
TC-10	FR-4	A user opens View Location on an offline session; then opens an online session.	An interactive map centred on the session’s coordinates is shown with pan/zoom and an “Open in Google Maps” link; the online session shows the meeting link instead of a map.	Pass

Test	FR	Scenario (planned)	Expected result (from design)	Actual result
TC-11	FR-5	A joined participant sends a message in the session chat room.	All participants in the room receive it within 2 seconds.	Pass
TC-12	FR-5	A user who has not joined tries to open or send messages in the chat room.	Access is denied (“Join this room to use its chat.”).	Pass
TC-13	FR-6	A user tries to join after the end of the registration deadline day or after the session start.	The session is shown as Closed and the join is rejected (“Registration is closed for this room.”).	Pass
TC-14	FR-7	A joined participant taps Add to Calendar.	Google Calendar opens with title, start/end time, location or meeting link and description pre-filled; 2 taps or fewer in total.	Pass
TC-15	NFR-5	Modify another account’s session, and fetch the chat messages of an unjoined session, directly through the database API.	Both attempts are rejected by the database security rules.	Pass

6. AI-Use

NUDTEE/Docs/AI use log.pdf at main · 6931503074-sudo/NUDTEE

7. Individual Contributions

Member responsibilities

Student ID	Name	Responsibility	links
6931503050	Ninmanee Nualsee	Design / Front / UI, Backend / data	https://github.com/6931503074-sudo/NUDTEE1/commits/main/?author=Ninmanee
6931503065	Rawisara Palang	Design / Front / UI, QA / AI log	https://github.com/6931503074-sudo/NUDTEE1/commits/main/?author=6931503065-web
6931503119	Suratsawadee Areerob	Documentation (SRS, diagrams), QA / AI log	https://github.com/6931503074-sudo/NUDTEE1/commits/main/?author=sSuratreerob
6931503074	Sirawit Kansuwan	Product / project lead (scope, planning), Documentation (SRS)	https://github.com/6931503074-sudo/NUDTEE1/commits/main/?author=6931503074-sudo

Student ID	Name	Responsibility	links
6931503107	Thanyarat Ainpasat	Documentation (SRS, diagrams	https://github.com/6931503074-sudo/NUDTEE1/commits/main/?author=6931503107-sketch

8. Golden Thread Traceability Table

Problem ⟶ Requirement ⟶ Design ⟶ Feature ⟶ Test

Problem	Requirement	Design	Feature (screen)	Test
No structured place to announce activities; details and locations get buried in chat threads.	FR-1	Session form with place search and current-location picker, online option, player limits and registration deadline validation; session stored in Cloud Firestore.	Create Session form with location selection, category and registration deadline controls. Status: Pass	TC-1, TC-2, TC-3. Result:Pass

Problem	Requirement	Design	Feature (screen)	Test
No reliable running headcount; organizers ask and count manually in chat.	FR-2	Cloud Firestore transaction for join/leave with capacity checking.	Join/Leave button with live attendee count (current/max). Status: Pass	TC-4, TC-5, TC-6, TC-7. Result:Pass
Headcounts go stale between chat messages as people change their minds.	FR-3	Cloud Firestore real-time listeners on session records.	Live-updating session list with status indicators (Open, Almost Full, Full, Joined, Closed). Status: Pass	TC-8, TC-9. Result:Pass
Unclear activity locations cause participants to get lost or arrive late.	FR-4	Embedded Google Maps view centred on the saved coordinates, with an “Open in Google Maps” link.	View Location screen and map preview on the session details. Status: Pass	TC-10. Result:Pass
Disorganised logistical coordination across general group chats.	FR-5	Per-session message collection in Cloud Firestore, opened only for participants.	Room Chat for joined attendees. Status: Pass	TC-11, TC-12, TC-15. Result:Pass

Problem	Requirement	Design	Feature (screen)	Test
Uncontrolled last-minute joins after event preparations are finalised.	FR-6	Time check (end of deadline day and session start) in the join flow before the join transaction.	Session shown as Closed and late join attempts rejected; ended sessions marked completed. Status: Pass	TC-13. Result:Pass
Participants forget scheduled activities for lack of reminders.	FR-7	Google Calendar event-template link pre-filled with the session details.	“Add to Calendar” button on the session details. Status: Pass	TC-14. Result:Pass

9. Relevant Links and References

Repository: https://github.com/6931503074-sudo/NUDTEE1

Architecture: NUDTEE/Docs/Architecture.md at main · 6931503074-sudo/NUDTEE

Traceability: NUDTEE/Docs/traceability.md at main · 6931503074-sudo/NUDTEE

Use Case diagram: NUDTEE/Docs/diagrams/Use Case Diagram - NUDTEE Application (1).pdf at main · 6931503074-sudo/NUDTEE

Sequence diagram: NUDTEE/Docs/diagrams/NUDTEE_SequenceDiagrams.pdf at main · 6931503074-sudo/NUDTEE

README: https://github.com/6931503074-sudo/NUDTEE1/blob/main/README.md

Prototype, demo:https://nudtee-e329d.web.app/